EV MARKET PERFORMANCE AND INSIGHTS

PROJECT REPORT

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Domain	Data Analytics and Business Intelligence
Tool	Tableau Public

A data-visualisation project exploring EV sales, manufacturers, vehicle categories, battery technologies and driving-range characteristics.

1. INTRODUCTION

1.1 Project Overview

The electric-vehicle (EV) industry is an important and developing part of the transportation sector. EV manufacturers offer different vehicle types and use different battery technologies, capacities and designs. Understanding this market requires examining several factors together, including sales, vehicle categories, manufacturers, manufacturing countries, customer ratings and driving range.

The project **EV Market Performance and Insights** uses a structured dataset and Tableau Public to present these factors through visualisations. The aim is to turn tabular information into charts that are easier to read, compare and discuss. Nine visualisations are organised into a single dashboard. This is an educational analysis of the supplied dataset and should not be treated as a complete measurement of the worldwide EV market.

1.2 Purpose

The purpose is to demonstrate how data visualisation can support basic market analysis. The dashboard helps viewers identify patterns in the available records, compare groups and form questions for further investigation.

2. IDEATION PHASE

2.1 Problem Statement

EV information can be difficult to interpret when it is presented only as a large table. Comparing sales across years, categories and manufacturers, or examining battery chemistry and driving range, can take time. This project addresses the presentation problem by creating a dashboard that summarises selected fields in visual form.

2.2 Empathy Map Canvas

- **Thinks:** Which vehicle categories and manufacturers have stronger sales in this dataset?
- **Wants to know:** How do battery technologies, battery capacity and driving range differ?
- **Needs:** Clear titles, readable labels, comparable values and a single place to view the analysis.
- **Challenges:** Interpreting many rows and understanding the meaning of different measures.

These points are a design exercise and are not the result of formal user interviews.

2.3 Brainstorming

- **Market performance:** annual sales trend, sales by vehicle category and sales by manufacturer.
- **Customer and geography:** average customer rating by category and sales by manufacturing country.
- **Technology and product comparison:** battery chemistry distribution, average driving range by category, battery capacity versus driving range, and sales by manufacturer and category.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

A typical user journey is: open the published dashboard, read the project title and chart headings, review the overall sales trend, compare categories and manufacturers, examine battery and range information, and use the combined views to form a general understanding of the dataset.

3.2 Solution Requirement

- A dataset containing EV-related fields suitable for comparison and aggregation.
- A visualisation tool capable of producing charts and combining them into a dashboard.
- Charts with meaningful titles, axes and suitable aggregation methods.
- A published dashboard link accessible to the intended reviewer.
- A report describing the project process, outputs, observations and limitations.

3.3 Data Flow Diagram

EV dataset (CSV)
↓
Connect and inspect data in Tableau
↓
Create aggregations and visualisations
↓
Arrange worksheets in a dashboard
↓
Publish using Tableau Public

3.4 Technology Stack

Component	Purpose
CSV dataset	Stores the EV records and attributes used in the analysis.
Tableau Public	Connects to data, creates charts and combines them into a dashboard.
Web browser	Opens the published visualisation.
PDF report	Documents the project approach, outputs and observations.

4. PROJECT DESIGN

4.1 Problem–Solution Fit

A tabular dataset does not immediately communicate comparisons and patterns. Tableau visualisations provide a suitable solution: line charts show change over time, bar charts compare categories, a scatter plot compares two numerical measures, and a heatmap shows a two-dimensional comparison.

4.2 Proposed Solution

The proposed solution is an interactive dashboard titled **EV Market Performance & Insights**. It contains nine views: annual sales trend, customer rating by category, sales by manufacturing country, sales by vehicle category, battery chemistry distribution, average driving range by category, sales by manufacturer, battery capacity versus driving range, and sales by manufacturer and category.

4.3 Solution Architecture

The CSV dataset is connected to Tableau Public. Fields are identified as dimensions or measures, and suitable aggregations are selected for each chart. Worksheets are arranged on a dashboard and published to Tableau Public. Viewers can access the published dashboard through its web link.

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

Stage	Activity	Output
Data preparation	Inspect the supplied EV dataset and identify fields.	Dataset ready for analysis.
Visualisation	Create worksheets for nine analytical questions.	Nine chart worksheets.
Dashboard design	Arrange worksheets and add a project heading.	Combined dashboard.
Publishing	Publish workbook and check public page.	Shareable dashboard link.
Documentation	Prepare report, screenshots and final links.	Project report.

These stages describe the workflow. Exact dates and durations can be added if required by the course template.

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

The dashboard was reviewed visually to confirm that the worksheets were present, chart titles were understandable and the views could be displayed together. Before final submission, the following checks should be completed:

- Open the published Tableau Public link and confirm that the dashboard loads.
- Check that all nine visualisations are visible by scrolling through the dashboard.
- Check chart titles, axis labels and category names for readability.
- Verify that numerical measures use the intended aggregation, such as SUM for sales and AVG for average ratings or driving range.
- Check the battery-capacity and driving-range scatter plot for the intended level of detail.
- Confirm that the dashboard is accessible to the reviewer.

Data-quality note: If customer rating is recorded on a scale such as 1–5, the chart should use AVG(Customer Rating), not SUM(Customer Rating). This report does not claim formal load-time or stress testing.

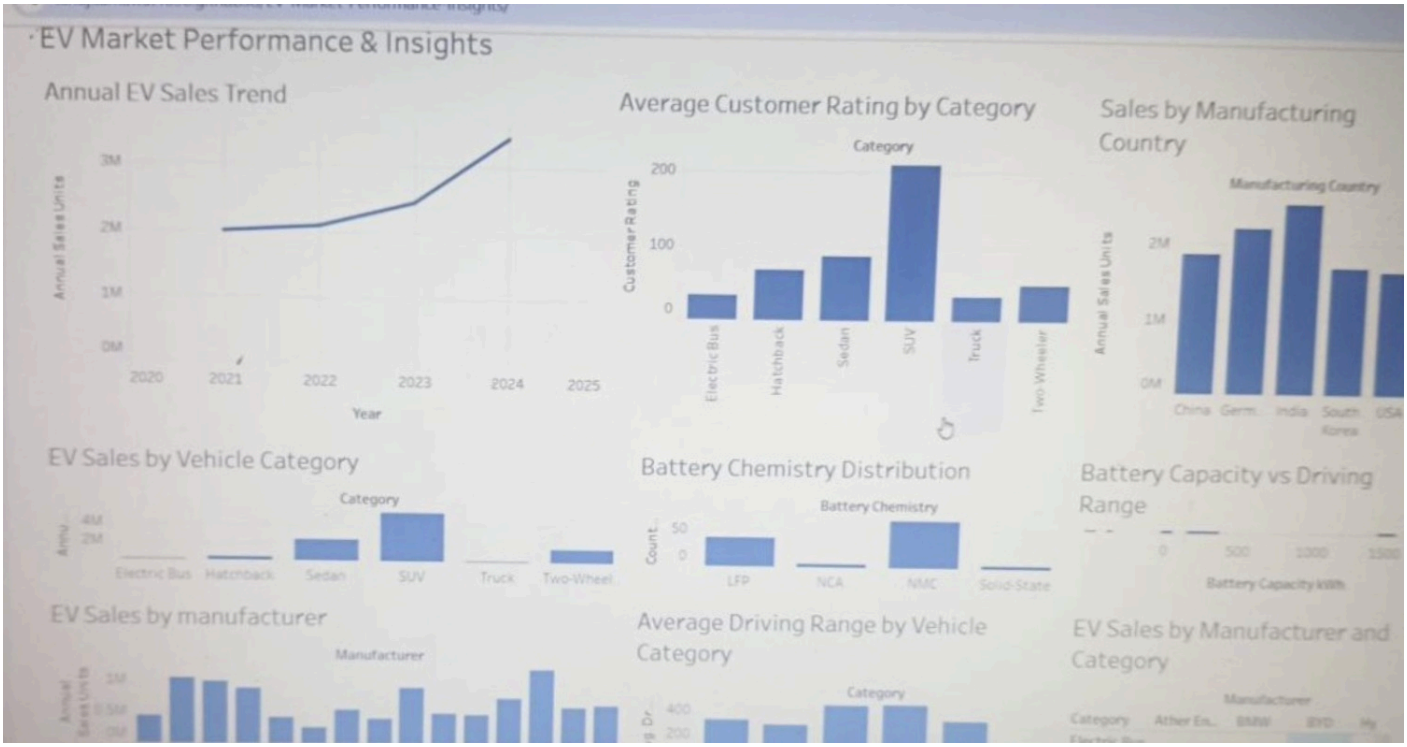
7. RESULTS

7.1 Output Screenshots

Insert screenshots of the final published dashboard and, if space permits, individual worksheets in this section. Suggested caption: *Figure 1: EV Market Performance & Insights dashboard published using Tableau Public.*

7.1 OUTPUT SCREENSHOT

Figure 1. EV Market Performance & Insights dashboard published online.



The dashboard combines nine visualisations covering EV sales, vehicle categories, manufacturers, manufacturing countries, battery chemistry, customer ratings and driving-range characteristics.

No.	Visualisation	What it communicates
1	Annual EV Sales Trend	Change in annual sales across the years in the dataset.
2	Average Customer Rating by Category	Average rating comparison across vehicle categories.
3	Sales by Manufacturing Country	Sales comparison across manufacturing countries.
4	EV Sales by Vehicle Category	Sales comparison across vehicle categories.
5	Battery Chemistry Distribution	Count of model records associated with each battery chemistry.
6	Average Driving Range by Vehicle Category	Average driving-range comparison across categories.
7	EV Sales by Manufacturer	Sales comparison across manufacturers.
8	Battery Capacity vs Driving Range	Relationship between battery capacity and driving range.
9	EV Sales by Manufacturer and Category	Colour-based comparison across manufacturers and categories.

8. ADVANTAGES & DISADVANTAGES

8.1 Advantages

- Presents several EV market measures in one dashboard.
- Makes comparisons across manufacturers, countries and vehicle categories easier.
- Uses different chart types for different analytical questions.
- Can be opened online through a published Tableau Public link.
- Helps viewers identify patterns and areas for further investigation.

8.2 Disadvantages and Limitations

- The analysis is limited to the fields and records in the supplied dataset.
- The dataset may not represent every EV manufacturer, country, model or year.
- The dashboard provides descriptive analysis and does not prove why a trend occurred.
- Conclusions depend on the accuracy, completeness and consistency of the source data.
- A dashboard with many charts may require scrolling or a larger screen.

9. CONCLUSION

This project demonstrates how Tableau Public can convert structured electric-vehicle data into visual information. The nine visualisations provide different views of sales, vehicle categories, manufacturers, manufacturing countries, battery chemistry, customer ratings, battery capacity and driving range. The dashboard makes it easier to compare these attributes and communicate observations. The project also provided practical experience in working with a dataset, selecting chart types, applying aggregations, arranging a dashboard and publishing a visualisation online.

10. FUTURE SCOPE

- Add a larger and more frequently updated dataset from reliable sources.
- Add filters for year, country, manufacturer and vehicle category.
- Include charging time, battery life, emissions and total cost of ownership if reliable data is available.
- Explore forecasting after obtaining sufficient time-series data and validating the model.
- Improve the layout for mobile screens and add clear KPI summaries.
- Add source notes and data-refresh information.

11. APPENDIX

11.1 Source Code

This is primarily a Tableau Public data-visualisation project. If no separate programming source-code file was created, write: *Not applicable – no separate programming source code was developed for this Tableau dashboard.*

11.2 Dataset Link

Insert the shareable Google Drive link to the dataset used in Tableau. If the reviewer needs access, set General access to “Anyone with the link – Viewer”.

11.3 GitHub & Project Demo Link

Tableau Public dashboard: Open EV dashboard

GitHub repository: Add your own repository URL after confirming that it contains your project files. Do not use another person's repository as your own.

Project demonstration video: Add the shareable link after uploading the final recording.

Dataset link: Add the public-view link to the dataset file used for this analysis.

FINAL SUBMISSION CHECK

Before submitting, add the dataset, GitHub and video links; insert actual output screenshots in Section 7; verify the customer-rating aggregation and other chart calculations in Tableau; and confirm that the published dashboard is accessible to the reviewer.

End of Report